

Context length and readout parameterisation select the emergence exponent of in-context skills in an additive model

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Abstract

We study how a feature-learning model acquires *skills* that are presented in context. Targets are additive models $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$ with mean-zero task coefficients c_p , and a student with learned features and a linear-in-label attention readout must recover the directions v_p by online SGD from prompts of length N . For a single skill of information exponent k^* we obtain the population loss in closed form. As $N \rightarrow \infty$ it is the in-weight loss with the link squared (effective exponent $2k^*$, a fact implicit in one-step analyses that we make dynamical). At finite N the loss acquires an *alignment-dependent* variance term $\Gamma^2 V(m)/N$, and the way the readout Γ is parameterised decides what this term does: a readout pinned at order one makes the unaligned state drift-stable below m^* , so that more prompts cannot substitute for context length; a free scalar readout shrinks to a Wiener factor: for the population flow at fixed context length the exponent becomes $4k^*$ with escape time $\propto 1/N$ (Theorem B), while at accessible d online SGD reproduces the flow’s intermediate slowdown, set jointly by context length and the readout’s initialisation and learning rate; a readout tied to the feature norm removes the N -dependence as long as its initial norm is below an explicit threshold (proved, with the N -dependence bounded by $k^* \rho_0 / (Nm_0^{2k^*-2})$). A parameter-free population ODE reproduces the exponents measured by online SGD at $d = 32$ for three readout protocols and four context lengths (all 12 cells within 0.5 by a censored fit), the escape times at $d = 64$ and $d = 128$ including the predicted flip of a context length from escape to trap, and, at $k^* = 3$, the trap threshold and an exponent near 6 (two pre-registered precision bands were missed and are reported). With

many skills the dynamics decouple with emergence times $\propto 1/\pi_p$; additive compositions of learned skills follow by linearity. In small softmax transformers no parameter whose scale we froze reproduces the trap or the rescue (two pre-registered tests, both killed), and, unlike in the solvable model, prompts per step partly substitute for context length.

1. Introduction

Theories of feature learning explain *when* a network acquires a direction v in input space in terms of the information exponent k^* of the link function (Ben Arous et al., 2021): online SGD needs d^{k^*-1} samples ($d \log d$ for $k^* = 2$). Large models, however, acquire much of their behaviour as *in-context* skills: the pretraining data presents a skill through prompts whose labels are generated by a task-dependent coefficient, and the network must learn features that make the skill recoverable from a context of length N . Recent analyses of in-context learning with learned features (Oko et al., 2024; Nishikawa et al., 2025; Kim & Suzuki, 2024; Gu et al., 2026) treat the context length as an *inference-time* resource: it controls how well a trained model estimates the task, and in one-step analyses of pretraining it enters only through concentration, so that more prompts can substitute for longer prompts (Oko et al., 2024, Remark 3).

We ask a different question: *does the context length, together with how the readout is parameterised, change which exponent governs the acquisition of a skill during pretraining?* We answer it in the smallest setting in which the question is sharp—one or several skills with $k^* = 2$ (even links, where in-weight learning is easy), a student with features on the sphere and a linear-in-label readout, online SGD—and then test the predictions outside the solvable model.

2. Setting

Inputs $x \sim \mathcal{N}(0, I_d)$. A *skill* is a unit vector $v_p \in S^{d-1}$ with link σ ; we use the normalised Hermite polynomials $\sigma_k = \text{He}_k / \sqrt{k!}$, so $\mathbb{E}[\sigma_j(\langle u, x \rangle) \sigma_k(\langle w, x \rangle)] = \delta_{jk} \langle u, w \rangle^k$ for unit u, w and $k^* = k$. A *task* is a coefficient vector c with independent mean-zero entries, $\mathbb{E}c_p^2 = s_p^2$; a *prompt*

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consists of N pairs (x_i, y_i) with $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$ and a query (x_q, y_q) from the same task.

Model A (in-context). Features $\phi_j(x) = \sigma(\langle w_j, x \rangle)$, $w_j \in S^{d-1}$, and a linear-in-label readout

$$\hat{y}_q = \sum_j \Gamma_j \phi_j(x_q) A_j, \quad A_j = \frac{1}{N} \sum_{i=1}^N y_i \phi_j(x_i), \quad (1)$$

the MLP-plus-linear-attention architecture of [Oko et al. \(2024\)](#). We consider three readout protocols: *fixed* $\Gamma_j = \gamma$; *free*, a trainable scalar; and *tied*, $\Gamma_j = \|u_j\|^2$ with $w_j = u_j/\|u_j\|$ (the 2-homogeneous student of [Ren et al. \(2025\)](#)). **Model B (in-weight)** is ordinary regression $\hat{y} = \sum_j a_j \phi_j(x)$ on the fixed task $c \equiv a$. Both are trained by online spherical SGD on the square loss with fresh data at every step: B prompts per step, step size η on the features (renormalised to the sphere after each step), step size η_Γ on a free readout initialised at Γ_0 , and for the tied readout an unnormalised u with $\|u_0\|^2 = \rho_0$. We report escape times $T_{0.5}$ (steps until $|m| \geq \frac{1}{2}$) and their *flow-time* version $T_{0.5} \eta$, which is what the population ODEs predict; the single-feature experiments initialise at $m_0 = c d^{-1/2}$ exactly.

Write $g(m) = \sum_k \alpha_k^2 m^k$ for $\sigma = \sum_k \alpha_k \sigma_k$ and $V(m) = \mathbb{E}[\sigma(\langle v, x \rangle)^2 \sigma(\langle w, x \rangle)^2] = \sum_j \beta_j^2 m^j$ where $\sigma^2 = \sum_j \beta_j \sigma_j$. For σ_2 : $g = m^2$, $V = 1 + 8m^2 + 6m^4$.

3. A single skill

Proposition 1 (Population loss). *For $P = 1$, one feature at overlap $m = \langle w, v \rangle$, readout Γ and $\mathbb{E}c^2 = s^2$,*

$$\begin{aligned} L_A(m) &= s^2 \left[1 - 2\Gamma g(m)^2 + \Gamma^2 g(1) \left(\left(1 - \frac{1}{N}\right) g(m)^2 + \frac{V(m)}{N} \right) \right], \\ L_B(m) &= 1 - 2a g(m) + a^2 g(1). \end{aligned} \quad (2)$$

(Proof: Appendix.)

The link is squared. As $N \rightarrow \infty$, $L_A = s^2[1 - 2\Gamma g^2 + \Gamma^2 g(1)g^2]$ depends on m only through $g(m)^2$, whereas L_B depends on it through $g(m)$. Near $m = 0$ the drift is $2\Gamma s^2(2 - \Gamma g(1)) k^* \alpha_{k^*}^4 m^{2k^*-1}$ against $2ak^* \alpha_{k^*}^2 m^{k^*-1}$ for Model B: the label enters the prediction twice—through the context statistic and through the comparison with y_q —and the mean-zero task prior removes the cross terms that would provide a degree- k^* path.

Proposition 2 (Finite context: a drift-stable unaligned state). *For σ_2 and fixed readout $\Gamma = \gamma$, $-\partial_m L_A = s^2[(8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N]$. For $\gamma < \gamma_1 = 2N/(N+9)$ the origin is a local minimum of L_A , the drift is repulsive for $m < m^*$ and attractive on $(m^*, 1)$, with*

$m^{*2} = 4\gamma/(2N - (N+5)\gamma) \approx 2\gamma/(N(1 - \gamma/2))$ (for $\gamma \geq \gamma_1$ the overlap decays from every $m_0 < 1$).

A feature initialised at $m_0 \approx d^{-1/2}$ is attracted only if $d \lesssim d^* = N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N)/(16\gamma)$ (≈ 31 for $\gamma = 1$, $N = 128$).

Proposition 3 (Readout parameterisation: three regimes—informal summary of Theorems A–B and Prop. 4, Appendix B). *Consider the population gradient flow of (2) for $\sigma = \sigma_2$ in flow time $t = \eta \cdot \text{steps}$, started at $m_0 = d^{-1/2}$, and let τ_{esc} be the time to reach $m = \frac{1}{2}$.*

(a) Pinned readout $\Gamma = \gamma$. *If $m_0 < m^*(\gamma, N)$ then $\tau_{\text{esc}} = \infty$; otherwise $\tau_{\text{esc}} = \Theta(m_0^{-2})$ for m_0 bounded away from m^* , diverging logarithmically as $m_0 \downarrow m^*$ (Theorem A(ii)), with an N -dependence that saturates for $N \gg \gamma d$. The condition involves N and γ only—not the number of prompts per step, which does not enter the population drift.*

(b) Free scalar readout $\dot{\Gamma} = -r \partial_\Gamma L_A$, $r = \eta_\Gamma/\eta$, $\Gamma(0) = \Gamma_0$. *The fixed point of Γ at given m is $\Gamma^*(m) = g^2/(g(1)[(1 - \frac{1}{N})g^2 + V/N])$, approached at rate $2r g(1)[(1 - \frac{1}{N})g^2 + V/N] \approx 2r/N$. Adiabatic case ($2r/N \gg$ the overlap rate): the overlap sees $L_A(m, \Gamma^*)$, whose exponent is $4k^*$ for $m^{2k^*} \ll V/N$ and $2k^*$ otherwise, and in the former regime $\tau_{\text{esc}} \propto m_0^{-(4k^*-2)}/N$ (Appendix A). Growth-limited case ($2r/N \lesssim$ the overlap rate; E2 at $N = 16$ is adiabatic to 1%, the other cells are not): Γ lags Γ^* and τ_{esc} depends on (N, Γ_0, r) jointly; the ODE gives intermediate N -dependences (e.g. $\tau(16)/\tau(256) = 4.4$ at $d = 64$, $\Gamma_0 = 0.01$, $r = 1$, against 16 in the adiabatic $1/N$ law).*

Tied readout $\Gamma = \rho = \|u\|^2$, features $\sigma(\langle u/\|u\|, x \rangle)$. *The directional velocity is $(1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$, which is exactly the pinned velocity at $\gamma = \rho$ divided by ρ (norm-independence of the attraction, [Ren et al., 2025](#), Lemma B.1): a tied readout behaves like a pinned readout of value $\rho(t)$ under a time change. From a small ρ_0 , $\dot{\rho} = -4\rho \partial_\rho L_A$ leaves ρ near ρ_0 and $\tau_{\text{esc}} = \Theta(m_0^{-2})$ is independent of N provided $\rho_0 \leq \rho^*(m_0)/4$ (Theorem A(iii)(c): $\rho(t)$ then stays within explicit factors of ρ_0 and $|\tau_{\text{esc}}(2k^* - 2)4k^* m_0^{2k^*-2} - 1| \leq C_k(k^* \rho_0/(N m_0^{2k^*-2}) + \rho_0 + \lambda_k m_0^2)$, $C_2 = C_3 = 4$; the qualifier was added after E6, and the two cells where N -dependence appeared—E5's $\rho_0 = 0.1$, $N = 16$ and E6's $N = 16$, $d = 16$ —are exactly those violating the hypothesis—within it the tied readout self-selects the small-readout regime. Let $\rho^*(m) = 2Nm^2/(4 + (N+5)m^2) \approx Nm^2/2$ be the value at which the directional drift vanishes. For $\rho_0 > \rho_{\text{trap}}(m_0)$ (Theorem A(iii); $\rho_{\text{trap}} \rightarrow c_2 \rho^*$ with*

$c_2 = \frac{4}{3}$ as $m_0 \rightarrow 0$, Remark 1; at $d = 64$, $N = 16$, $\rho_{\text{trap}} = 0.178$ against $c_2\rho^* = 0.154$; between ρ^* and ρ_{trap} the overlap first falls and then escapes, proved for the small- m system), the population flow traps permanently ($m \propto t^{-2}$ while $\rho \propto t^{-1}$, so ρ never falls below the moving threshold); under SGD, noise keeps m at the noise floor while ρ decays, and the feature is eventually released—a long delay rather than a trap (E5).

That a fast readout doubles the exponent of an in-weight problem is known (Bietti et al., 2023, §5); what is specific to the in-context problem is that the finite-context noise drives the readout towards Γ^* , so that *whether context length trades off against training time, is irrelevant, or is a hard requirement is decided by how the readout is parameterised.*

E1: exponents by SGD. Fig. 2 (Appendix) shows κ_{eff} for free ($\eta_{\Gamma} = \eta$ and 10η) and tied readouts at $N \in \{32, 128, 512, 4096\}$ against the population ODE of Props. 1–3 evaluated *after the run* at the same d , m_0 -grid, Γ_0 and η_{Γ}/η (the ODE itself was fixed before the run; its evaluation at these parameters was not pre-registered): tied 4.10–4.44 (ODE 4.15–4.18); free 4.8–6.6 rising as N falls (ODE 4.85–7.0). With a censored (Tobit) fit for the three cells that contain censored runs, all twelve cells agree with the ODE within 0.5 (ordinary least squares on reached runs only: eleven within 0.6, the free $N = 32$, $\eta_{\Gamma} = \eta$ cell 1.0 low); the estimator is a secant over a factor 2.8 in m_0 and local slopes vary, so these are effective, not asymptotic, exponents.

E2 and E7: the three regimes by SGD at $d = 64$ and 128 (Exps. 6 and 10, P23). To test Prop. 3 directly (Fig. 1) we fix $d = 64$, $m_0 = d^{-1/2}$, $\eta = 1/d^2$, vary $N \in \{16, 64, 256\}$ under two schemes—tokens per step fixed ($NB = 4096$) and prompts per step fixed ($B = 64$)—and run the four protocols, three seeds each, with the population escape times computed before the run. Pinned $\gamma = 1$: no escape at $N \leq 64$ in 10^6 steps (0/9, including $B = 256$), 3/6 at $N = 256$ ($d^* \approx 63$); pinned $\gamma = 0.1$: median 396k/192k/179k steps against 400k/193k/175k predicted; free: 1.78M/614k/402k against 1.82M/623k/409k, ratio $T(16)/T(256) = 4.4$ (3.6–5.0 over seed pairings) as predicted (4.44)—the growth-limited value of Prop. 3(b), not the adiabatic $1/N$ law; tied: 17.0k/15.1k/14.0k against 17.0k/16.4k/16.3k (flat within $\times 1.25$, seed pairings 0.9–1.45). Repeating E2’s protocols at $d = 128$ ($\eta = 1/d^2$, three seeds): the pinned $\gamma = 0.1$ cell at $N = 16$, which escaped at $d = 64$, is now censored 3/3 at $2.5 \cdot 10^6$ steps—the flip across $d^*(16) = 76$ of Prop. 3(a)—while $N = 64, 256$ escape at 0.99 and 1.01 of the flow time (1.78 and $1.47 \cdot 10^6$ steps; ratio 1.22, N -saturating); pinned $\gamma = 1$ is censored at every N ($d^*(256) = 63$); the tied readout stays N -

flat (medians/flow 1.12, 0.97, 1.02; $T(16)/T(256) = 1.20$, flow 1.09); the free readout at $N = 256$ takes $5.29 \cdot 10^6$ steps (0.985 of the flow) and its secant exponent across $d = 64 \rightarrow 128$ is 5.4 (flow 5.4), between the signal-dominated 4 and the adiabatic 8, and close to the large- λ , small- Γ_0 corner $2k^* + 1 = 5$ of Prop. 4.

E5: controls. Varying the readout’s initial value and learning rate (free: $\Gamma_0 = 0.1$ or $\eta_{\Gamma} = 10\eta$ at $d = 64$) changes the N -ratio from 4.4 to 9.2 and 10.7 (ODE 9.1 and 8.9; medians within 2% except the $\eta_{\Gamma} = 10\eta$, $N = 16$ cell, 24% slow), confirming that the free-readout exponent is a function of $(N, \Gamma_0, \eta_{\Gamma}/\eta)$, as Prop. 3(b) states. A tied readout started at $\rho_0 = 0.3 > \rho_{\text{trap}}(m_0) = 0.18$ at $N = 16$ was pre-registered to trap (the deterministic flow provably does, Theorem A(iii): $m \propto t^{-2}$, $\rho \propto t^{-1}$); under SGD all three seeds escaped after a $\approx 30\times$ delay, once ρ had decayed to 0.012—so under SGD tying converts the pinned trap into a delay, and the pre-registered wording “the repulsion survives tying” is withdrawn in favour of “is delayed by tying”.

E6: a $k^* = 3$ check (Exp. 9, pre-registered P22). With $\sigma = \sigma_3$ ($g = m^3$, $V = 1 + 18m^2 + 54m^4 + 20m^6$), $\eta = 2 \cdot 10^{-4}$, $B = 64$, $d \in \{8, 16, 32\}$, three seeds: the tied readout ($\rho_0 = 0.01$, $N = 128$) gives medians 1.18/0.89/0.80 of the flow ($d = 8/16/32$; the pre-registered band was 15%, so P22a failed) and secant exponents 5.7 and 6.0 (ODE 6.5, 6.4; the $8 \rightarrow 16$ secant misses its ± 0.5 band)—an exponent near 6, against ≈ 4 at $k^* = 2$. The pre-registered N -flatness of the tied readout (P22b) *failed* at $N = 16$, $d = 16$: the flow predicts $1.6\times$ the $N = 128$ time, the seeds gave 47k, 500k and censored, and at $d = 32$ all three seeds trapped (predicted: $\rho_0 = 0.01 > \rho_{\text{trap}} = 0.006$). The statement that survives is the conditional one in Prop. 3(c): N -flat for $\rho_0 \ll \rho_{\text{trap}}(m_0, N)$.

4. Many skills and composition

E3 ($d = 32$, $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, fixed $\Gamma = 0.1$, three seeds) gives slopes of $\log T_p$ versus $\log p$ of 1.44 ± 0.08 for the in-context model, against 0.75 ± 0.08 for a capacity-matched in-weight baseline and 0.75 – 0.89 for the 2-homogeneous in-weight student of Ren et al. (2025). (Appendix)

5. Transformers: where the effect shows and where it does not

We train two-layer softmax transformers (4 heads, pre-LN, Adam, online data; Appendix, Fig. 5) on the same in-context task with $k^* = 2$ and ask whether a short context can be compensated by more prompts per step. **Large initial alignment** ($d = 32$, MLP width 256): at a fixed

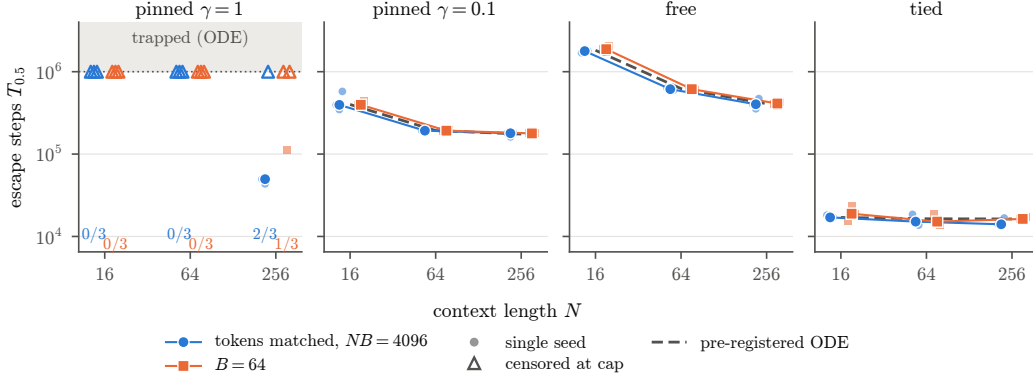


Figure 1. Escape steps $T_{0.5}$ (first step with $|m| \geq 0.5$) against context length N for the four readout protocols ($d = 64$, $\eta = 1/d^2$, $m_0 = d^{-1/2}$; shared log-log axes). Light markers are single seeds (three per cell), solid markers and lines the median over seeds, for tokens per step fixed ($NB = 4096$, circles) and prompts per step fixed ($B = 64$, squares). Open triangles are runs that had not escaped at the step cap (10^6) and are drawn at the cap; the counts under the pinned- $\gamma = 1$ panel are runs escaped out of 3. Dashed lines are the population-ODE escape times (flow time $\times d^2$), fixed before the run; pinned $\gamma = 1$ has no finite ODE escape time (trapped, shaded beyond the cap).

8192 tokens per step the emergence step *increases* with N (1000/1400/2800 for $N = 16/64/256$)—fewer prompts per step dominate and the mechanism is invisible, as it must be when $m_0 \gg m^*$. **Small initial alignment** ($d = 256$, width 32; $|\cos|_0 = 0.13\text{--}0.18$): $N = 16$ with $B = 64$ prompts never emerges in 30k steps (3/3); $N = 16$ with $B = 1024$ (the same tokens per step as $N = 256$, $B = 64$) emerges in all seeds but at 10.6–14.8k steps, $\approx 3\times$ later than $N = 256$ (2.8–4.0k). Neither pre-registered extreme (a trap, or interchangeability of N and B) holds; the $\approx 3\times$ ratio rests on two paired seeds, the Adam step was not rescaled with B . We state the result as an observation; it is not a test of the mechanism of Sec. 3, and in one respect—prompts per step rescue the short context under every protocol below—it differs from it. If Prop. 3 transferred through the final linear readout, freezing that readout at its $O(1)$ initial norm (pinned) should turn the $N = 16$ handicap into a trap that $B = 1024$ cannot lift, and freezing it at $0.1\times$ (pinned small) should rescue $N = 16$, $B = 64$. Neither happened: the $O(1)$ -frozen readout emerged at 5.8k and 8.4k steps with $B = 1024$ —no later than the trainable one—and the small frozen readout stayed on the plateau for 30k steps at $B = 64$ in both seeds, while $N = 256$ emerged under every protocol. A second pre-registered test (Exp. 11, P24) moved the knob to the attention output projections of both blocks, the path through which the context statistic enters the residual stream: frozen at their initial norm, $N = 16$ was again stuck at $B = 64$ (2/2) and rescued by $B = 1024$ (15.0k, 26.4k); frozen at $0.1\times$, $N = 16$, $B = 64$ stayed stuck (2/2). Both kill criteria were met. The three-regime picture is therefore established for the solvable model only. We make no transfer claim.

6. Related work

Oko et al. (2024) analyse the architecture (1) with one gradient step; the product of two correlations ($\propto m^{2Q-1}$) is in their Lemma 21 and the exponent $2Q$ in their total sample count $N_1 T_1 = \tilde{\Omega}(d^{2Q+1} r)$, framed as a consequence of correlational information; their Remark 3 states the N_1 – T_1 trade-off, which Prop. 3 recovers for a shrinking readout; a readout pinned at order one lies outside their $\gamma \propto d^{-Q}$ regime, and there the trade-off is absent (Prop. 2, Appendix Fig. 3)—a complement to, not a contradiction of, their Remark 3. Nishikawa et al. (2025) show that softmax’s label nonlinearity lowers the *inference-time* context length to the generative exponent while one-step pretraining stays at $d^{\Theta(\text{ie})}$. Gu et al. (2026) derive, by the replica method for trained models with $k^* = 1$, the kernel-vs-feature context-length scalings and the static shrinkage Γ^* (their eqs. (297)–(298)); they list jointly trained readouts and $k^* \geq 2$ as open, which is where our results sit. Kim & Suzuki (2024) note that at finite sample size the ICL landscape may acquire a spurious minimum at the origin (their App. C.4), a qualitative precursor of Prop. 2. Ren et al. (2025) treat the additive model in-weight for even links with $k^* > 2$ with a 2-homogeneous student; our tied protocol is theirs, the norm-independence of the directional dynamics is their Lemma B.1, and $k^* = 2$ is outside their assumptions.

Limitations. The theory is for one feature per skill (Theorems A–B hold for any $k^* \geq 2$; SGD checks are at $k^* = 2$ and 3); the SGD threshold is soft; the transformer results are observational, so no transfer claim is made.

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A. Derivations

Hermite algebra. For unit u, w with $\langle u, w \rangle = m$ and $z_u = \langle u, x \rangle$, $z_w = \langle w, x \rangle$ jointly Gaussian with correlation m , $\mathbb{E}[\sigma_j(z_u)\sigma_k(z_w)] = \delta_{jk}m^k$. For $\sigma = \sigma_2 = (z^2 - 1)/\sqrt{2}$: $\sigma_2^2 = \sqrt{6}\sigma_4 + 2\sqrt{2}\sigma_2 + 1$, hence $V(m) = \mathbb{E}[\sigma_2(z_u)^2\sigma_2(z_w)^2] = 6m^4 + 8m^2 + 1$ and $g(m) = m^2$. For $\sigma_1 = z$: $g = m$, $V = 1 + 2m^2$.

Population loss (Prop. 1). With $A = \frac{1}{N} \sum_i y_i \sigma(\langle w, x_i \rangle)$ and $y_i = c \sigma(\langle v, x_i \rangle)$, $\mathbb{E}[A | c] = c g(m)$ and $\text{Var}[A | c] = c^2(V(m) - g(m)^2)/N$. Since q is independent of the context, $\mathbb{E}[y_q \hat{y}_q] = \Gamma \mathbb{E}c^2 g(m)^2$ and $\mathbb{E}[\hat{y}_q^2] = \Gamma^2 g(1) \mathbb{E}c^2 (g^2 + (V - g^2)/N)$, giving (2).

Drift (Prop. 2). For σ_2 , $\Gamma = \gamma$, $s^2 = 1$: $L_A = 1 - 2\gamma m^4 + \gamma^2[(1 - \frac{1}{N})m^4 + (1 + 8m^2 + 6m^4)/N]$, so $-\partial_m L_A = (8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N$, which is negative for $m < m^*$ with $m^{*2} = 16\gamma/(N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N))$. (An earlier draft of this derivation dropped the $24\gamma^2/N$ term and had a factor 2 in the cubic coefficient; the closed-form loss, which the Monte-Carlo drift check verifies directly, was unaffected.)

General σ_k . The normalised second Hermite coefficient of σ_k^2 is $\beta_2 = \sqrt{2}k$ (checked by quadrature for $k \leq 4$), so the finite-context repulsion is $-\Gamma^2 \beta_2^2 2m/N = -4k^2 \Gamma^2 m/N$ to leading order while the attraction is $2\Gamma(2 - \Gamma g(1))k m^{2k-1}$: the origin is drift-stable below $m^{*2(k-1)} \approx 2k\Gamma/((2 - \Gamma)N)$ for every $k \geq 2$ (for $k = 1$ the two terms have the same power and the repulsion is a constant fraction). At $k = 3$, $\Gamma = 0.1$, $N = 128$ the trap sets in at $d^* \approx 17$ (against ≈ 607 at $k = 2$), consistent with the negative Monte-Carlo drift observed for $k^* = 3$ at $d = 16$. The step size $\eta \propto d^{-2}$ is the critical scaling only for $k^* = 2$.

Tied = pinned up to a time change. The tied directional velocity equals the pinned velocity at $\gamma = \rho$ divided by ρ ; with $\dot{\rho} = -4\rho \partial_\rho L_A$ the pair (m, ρ) above threshold obeys $m \propto t^{-2}$, $\rho \propto t^{-1}$ (LSODA to $t = 10^6$), and the moving threshold $\rho^*(m) \approx Nm^2/2 \propto t^{-4}$ falls faster than ρ : the population flow traps permanently.

Escape-time scalings in Prop. 3. (a) With $\Gamma = \gamma$ fixed and $m_0 > m^*$ the drift near the origin is $\approx c_\gamma m^{2k^*-1}$ with $c_\gamma = \gamma(8 - 4\gamma)k^* \alpha_{k^*}^4/2$ for $k^* = 2$, so $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}/c_\gamma$ up to the N -dependent correction $-16\gamma^2 m/N$, which saturates as $N \rightarrow \infty$. (b) Along Γ^* , for $g^2 \ll V/N$, $L_A(m, \Gamma^*) \approx s^2[1 - Ng(m)^4/(g(1)V(m))]$ and $V(m) \rightarrow \beta_0^2 = g(1)^2$ as $m \rightarrow 0$, so the drift is $\approx s^2 N 4k^* \alpha_{k^*}^8 m^{4k^*-1}/g(1)^3$: $\tau_{\text{esc}} \asymp m_0^{-(4k^*-2)}/N$, i.e. $\tau_{\text{esc}} \propto d^{2k^*-1}/N$ —the multiplicative N - τ trade-off. The condition for this regime at the initial overlap is $m_0^{2k^*} \ll V/N$, i.e. $d^{k^*} \gg N$; for $d^{k^*} \ll N$ the signal-dominated form $1 - g^2/g(1)$ applies and $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}$ is N -independent. (c) For the tied readout the attraction term $4gg'$ carries no ρ , so $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}/(4k^* \alpha_{k^*}^4)$ independently of N as long as $\rho V'/N \ll 4gg'$ along the trajectory, which holds from $\rho_0 \leq \rho^*(m_0)/4$ (Theorem A(iii)(c)), with the N -dependence bounded by $C_k k^* \rho_0/(Nm_0^{2k^*-2})$, which for $k^* = 2$ reads $\rho_0 \ll Nm_0^2/2$; for $k^* = 3$, $\rho_{\text{trap}} \propto m_0^4$ and $\rho_0 = 0.01$ already violates it at $N = 16$, $d \geq 16$ (Exp. 9). At $d = 512$, $k^* = 2$ the three regimes give $\tau \propto \{\infty \text{ or } N\text{-saturating, } N^{-1}, N^0\}$, matching the integrated flows ($\infty/745/379$; $8.9 \cdot 10^4/2.2 \cdot 10^4/5.3 \cdot 10^3$; $39/34/33$ for $N = 32/128/512$).

Population ODEs (Prop. 3). Spherical gradient flow for the overlap and plain flow for the readout: free readout $\dot{m} = -(1 - m^2)\partial_m L_A$, $\dot{\Gamma} = -r \partial_\Gamma L_A$ with $r = \eta_\Gamma/\eta$; tied readout $\dot{m} = (1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$, $\dot{\rho} = -4\rho \partial_\rho L_A$. The chain rule through $\hat{u} = u/\|u\|$ gives the directional velocity a factor $1/\|u\|^2$ that cancels the $\rho = \|u\|^2$ carried by the attraction term but not the ρ carried by the repulsion. Escape times $T_{0.5}$ to $m = 0.5$ from $m_0 = c d^{-1/2}$ were integrated with an adaptive step (1% relative change per step) and the exponent read off as $\kappa_{\text{eff}} = 2 - \partial \log T / \partial \log m_0$ over $c \in \{0.5, 0.7, 1, 1.4\}$, exactly as for the SGD data.

Kramers estimate. $D_{\text{eff}} = \eta \text{sd}^2/2$ with the per-step standard deviation of the m -component of the batch gradient measured by Monte Carlo (0.33–0.37 at $B = 32$); the estimate ignores the spherical correction term, which is of the same order as the quoted $d = 32$ ratio, so only the $d = 64$ value (≈ 20) is meaningful.

Monte-Carlo drift check. $d = 16$, $N = 128$, $\gamma = 0.1$, $a = 1$, $5 \cdot 10^4$ batches of 64 prompts per point. Ratios MC/formula at $m = 0.2$ and 0.4 : Model B $k^* = 1, 2, 3$: 0.998–1.00; Model A $k^* = 1$: 1.000; $k^* = 2$: 0.972, 0.998; $k^* = 3$: 0.88 ± 0.76 (drift ≈ 0), 1.02. Model A $k^* = 3$ at $\gamma = 0.1$, $N = 128$ has negative drift for $m \leq 0.2$.

B. Theorems and proofs

All closed forms in this section are checked numerically by `scripts/verify_theorems.py` (61 checks: thresholds, the exact escape time against LSODA, the time-change identity, the t^{-2}/t^{-1} asymptotics, Theorem B at d up to $6.5 \cdot 10^4$, the corner exponents).

Standing notation. $\sigma = \sigma_k$, $k \geq 2$, so $\alpha_k = 1$, $g(m) = m^k$, $g(1) = 1$. Write $\sigma_k^2 = \sum_{i=0}^k \beta_{2i} \sigma_{2i}$; the linearisation formula $\text{He}_k^2 = \sum_j j! \binom{k}{j} \text{He}_{2k-2j}$ gives $\beta_{2i}^2 = ((k-i)! \binom{k}{i})^2 (2i)!/k!^2 \geq 0$, in particular $\beta_0^2 = 1$, $\beta_2^2 = 2k^2$, $\beta_{2k}^2 = \binom{2k}{k}$, and $V(m) = \sum_{i=0}^k \beta_{2i}^2 m^{2i}$, $V'(1) = 4k \mathbb{E}[\sigma_k^2 \sigma_{k-1}^2] (= 40 \text{ for } k = 2)$. Put $G = g^2 = m^{2k}$, $D(m) = (1 - \frac{1}{N})m^{2k} + V(m)/N$, so that

$$L(m, \Gamma) = s^2[1 - 2\Gamma G + \Gamma^2 D], \quad \partial_m L = s^2(\Gamma^2 D' - 2\Gamma G'), \quad \partial_\Gamma L = 2s^2(\Gamma D - G), \quad \Gamma^*(m) = G/D.$$

Two elementary facts are used throughout: $D \geq V(0)/N = 1/N$, and $mV' \leq 2kV$ (hence $mD' \leq 2kD$), because V has degree $2k$ and nonnegative coefficients. Flows: pinned $\dot{m} = -(1 - m^2)\partial_m L(m, \gamma)$; free $\dot{m} = -(1 - m^2)\partial_m L$, $\dot{\Gamma} = -r\partial_\Gamma L$; tied $\dot{m} = -(1 - m^2)\partial_m L(m, \rho)/\rho$, $\dot{\rho} = -4\rho\partial_\rho L(m, \rho)$. The interval $(0, 1)$ is invariant for m in all three ($g'(0) = V'(0) = 0$ for $k \geq 2$), $\tau_{\text{esc}} = \inf\{t : m(t) \geq \frac{1}{2}\}$, and t is flow time (η -steps).

Theorem A (Finite-context trap; population flow, $\sigma = \sigma_k$, $k \geq 2$). (i) Pinned readout $\Gamma \equiv \gamma > 0$. $\dot{m} = s^2\gamma(1 - m^2)mQ_\gamma(m^2)$ with

$$Q_\gamma(x) = 2k(2 - \gamma(1 - \frac{1}{N}))x^{k-1} - \frac{\gamma}{N} \sum_{i=1}^k 2i\beta_{2i}^2 x^{i-1}.$$

Let $\gamma_1 = 2N/(N - 1 + V'(1)/2k)$. If $\gamma < \gamma_1$, Q_γ has exactly one positive root x^* , and $x^* \in (0, 1)$; with $m^* = \sqrt{x^*}$, $m(t) \downarrow 0$ (exponentially, at rate $\rightarrow 4k^2 s^2 \gamma^2 / N$) if $m_0 < m^*$, and $m(t) \uparrow 1$ with $\tau_{\text{esc}} < \infty$ if $m^* < m_0 < \frac{1}{2}$. If $\gamma \geq \gamma_1$, $m(t) \rightarrow 0$ from every $m_0 \in (0, 1)$. As $N \rightarrow \infty$ at fixed $\gamma \in (0, 2)$, $m^{*2(k-1)} = \frac{2k\gamma}{(2-\gamma)N} (1 + O(N^{-1/(k-1)}))$. For $k = 2$: $\gamma_1 = \frac{2N}{N+9}$ and $m^{*2} = \frac{4\gamma}{2N-(N+5)\gamma}$.

(ii) Exact escape time, $k = 2$. Let $C = \gamma(8 - 4\gamma - 20\gamma/N)$, $B = 16\gamma^2/N$ ($C > B \iff \gamma < \gamma_1$; the constants in the $O(\cdot)$ terms below are uniform for $\gamma \leq (1 - \delta)\gamma_1$, $\delta > 0$) and $m_0^{*2} < m_0^2 < \frac{1}{4}$. Then

$$\tau_{\text{esc}} = \frac{F(\frac{1}{4}) - F(m_0^2)}{2s^2}, \quad F(u) = -\frac{\ln u}{B} - \frac{\ln(1-u)}{C-B} + \frac{C \ln(Cu-B)}{B(C-B)}.$$

Hence $2s^2 C m_0^2 \tau_{\text{esc}} = 1 + O(m_0^2 \ln \frac{1}{m_0} + m^{*2}/m_0^2)$, and $\tau_{\text{esc}} \sim \frac{C}{2s^2 B(C-B)} \ln \frac{1}{m_0^2 - m^{*2}}$ as $m_0 \downarrow m^*$.

(iii) Tied readout. (a) $\dot{m}_{\text{tied}}(m, \rho) = \dot{m}_{\text{pinned}}(m; \gamma=\rho)/\rho$ identically. Hence, if $(m, \rho)(t')$ is the tied solution and $t = \int_0^{t'} du/\rho(u)$, then $m(t)$ solves the pinned flow with the time-dependent value $\gamma(t) = \rho$ (i.e. $dt' = \rho dt$); for ρ frozen at ρ_0 , $\tau_{\text{tied}} = \rho_0 \tau_{\text{pinned}}(\gamma=\rho_0)$. (b) Let $m_0 < \frac{1}{2}$, $D_0 = D(m_0)$ and $a = k^2(k-1)(1 - m_0^2)/(ND_0)$. If $a > 1$ and

$$\rho_0 > \rho_{\text{trap}}(m_0) := \frac{k(k-1)m_0^{2k-2}}{D_0(a-1)}, \quad (3)$$

the tied flow is trapped for ever: $m(t) \leq m_0$ for all t , $m(t) \rightarrow 0$, and, as $t \rightarrow \infty$, $\rho(t)t \rightarrow N/(8s^2)$ and $m(t)t^{k^2/2} \rightarrow c \in (0, \infty)$; for $k = 2$, $m \propto t^{-2}$, $\rho \propto t^{-1}$. Since $\rho^*(m) := 2G'/D' \asymp Nm^{2k-2}/k$, $\rho/\rho^*(m) \rightarrow \infty$: the threshold recedes faster than ρ decays. (c) (Tied readout: N -flat escape for small ρ_0 .) Let $m_0 \in (0, \frac{1}{2})$, $\rho_0 \leq \frac{1}{4}\rho^*(m_0)$, and put $\varepsilon = k\rho_0/(Nm_0^{2k-2})$, $X = s^2(2k-2)4km_0^{2k-2}$, $\bar{v}_k = V'(\frac{1}{2})/(2k^2)$, $c_1 = 2V(\frac{1}{2})/(k(k-1))$. Then m increases to $\frac{1}{2}$ in finite time τ_{tied} , $\rho/\rho^*(m) < \frac{1}{3}$ throughout, and for $t \leq \tau_{\text{tied}}$

$$\frac{\rho_0}{1 + c_1\rho_0/(Nm_0^{2k-2})} \leq \rho(t) \leq \rho_0 \left(\frac{1 - m_0^2}{1 - m(t)^2} \right)^{3/(2k)} \leq \left(\frac{4}{3} \right)^{3/(2k)} \rho_0. \quad (4)$$

With $\rho_{\min}, \rho_{\max}$ the extremes of ρ on $[0, \tau_{\text{tied}}]$ and $\ell_k = (2k-2)m_0^{2k-4} \int_{m_0}^{1/2} m^{3-2k} dm (= 2 \ln \frac{1}{2m_0}$ for $k = 2$, $\leq \frac{k-1}{k-2}$ for

$k \geq 3$,

$$X\tau_{\text{tied}} \geq (1 - (2m_0)^{2k-2}) \left(1 + \frac{(1-\frac{1}{N})\rho_{\min}}{2}\right) + (1 - (2m_0)^{4k-4}) \frac{k\rho_{\min}}{2Nm_0^{2k-2}}, \quad (5)$$

$$X\tau_{\text{tied}} \leq 1 + \frac{4}{3}\ell_k m_0^2 + \rho_{\max} + \bar{v}_k \frac{k\rho_{\max}}{Nm_0^{2k-2}},$$

hence $|X\tau_{\text{tied}} - 1| \leq C_k(\varepsilon + \rho_0 + \lambda_k m_0^2)$ with $C_k = \max\{4, (\frac{4}{3})^{3/(2k)}\bar{v}_k\}$, $\lambda_2 = 1 + \ln \frac{1}{2m_0}$, $\lambda_k = 1$ ($k \geq 3$). For $k = 2$: $c_1 = \frac{27}{8}$, $\bar{v}_2 = \frac{11}{8}$, $C_2 = 4$; for $k = 3$: $c_1 = \frac{49}{16}$, $\bar{v}_3 = \frac{65}{24}$, $C_3 = 4$. The context length enters (4)–(5) only through ε and $1 - \frac{1}{N}$, so for $\varepsilon \rightarrow 0$ the escape time is N -flat.

(iv) For every number B of prompts per step the expected minibatch gradient is ∇L ; the population flows above, and hence $m^*(\gamma, N, k)$, γ_1 and (ii)–(iii), do not depend on B .

Lemma A.1 (Monotone signal-to-noise). $R(m) = G^2/D$ is strictly increasing on $(0, 1]$, and $\rho^*(m) = 2G'/D'$ satisfies $\rho^* > \Gamma^*$ and is strictly increasing.

Proof. $mR'/R = 4k - mD'/D \geq 2k > 0$. $\rho^* > \Gamma^* \iff 2G'D > GD' \iff 4kD > mD'$. Finally $\rho^* = 4k/(2k(1 - \frac{1}{N}) + N^{-1} \sum_i 2i\beta_{2i}^2 m^{2i-2k})$ and each m^{2i-2k} , $i < k$, is strictly decreasing. $\square$

Proof of Theorem A. (i) Insert $G' = 2km^{2k-1}$ and $V'(m)/m = \sum_{i \geq 1} 2i\beta_{2i}^2 m^{2i-2}$ in $-\partial_m L$. The coefficients of Q_γ of degree $< k-1$ are strictly negative ($\beta_{2i}^2 > 0$), and $Q_\gamma(0) = -4k^2\gamma/N < 0$. By Descartes' rule Q_γ has at most one positive root; it has one iff the leading coefficient $2k[2 - \gamma(N-1 + \beta_{2k}^2)/N]$ is positive, and the root lies in $(0, 1)$ iff $Q_\gamma(1) = 2k[2 - \gamma(N-1 + V'(1)/2k)/N] > 0$, i.e. $\gamma < \gamma_1$ (this implies the former, as $V'(1)/2k > \beta_{2k}^2$). Then $Q_\gamma < 0$ on $(0, x^*)$ and > 0 on (x^*, ∞) ; if $\gamma \geq \gamma_1$, $Q_\gamma < 0$ on $(0, 1)$. The one-dimensional phase portrait follows: on $[m_0, \frac{1}{2}] \subset (m^*, 1)$, $\dot{m} \geq c > 0$; on $(0, m_0] \subset (0, m^*)$, $\dot{m}/m \leq -c' < 0$ and $\dot{m}/m \rightarrow s^2\gamma Q_\gamma(0)$. For the asymptotics, the root equation $2k(2 - \gamma(1 - \frac{1}{N}))x^{*k-1} = \frac{\gamma}{N} \sum_i 2i\beta_{2i}^2 x^{*i-1}$ forces $x^* \rightarrow 0$ and gives $x^{*k-1} = \frac{4k^2\gamma}{2k(2-\gamma)N}(1 + O(x^* + N^{-1}))$ with $x^* = O(N^{-1/(k-1)})$. For $k = 2$, $Q_\gamma(x) = (Cx - B)/\gamma$.

(ii) With $u = m^2$, $\dot{u} = 2s^2u(1-u)(Cu - B)$, and $\frac{1}{u(1-u)(Cu-B)} = -\frac{1}{Bu} + \frac{1}{(C-B)(1-u)} + \frac{C^2}{B(C-B)(Cu-B)}$; integrate from u_0 to $\frac{1}{4}$. Expanding F in $\varepsilon = B/(Cu_0)$ gives $2s^2(F(\frac{1}{4}) - F(u_0)) = \frac{1}{Cu_0}(1 + O(\varepsilon)) + O(C^{-1} \ln \frac{1}{u_0})$; the logarithmic singularity at $u_0 = B/C$ is the last term of F .

(iii)(a) $\partial_m L(m, \gamma) = s^2\gamma(\gamma D' - 2G')$, so $\dot{m}_{\text{pinned}}/\gamma = s^2(1 - m^2)(2G' - \gamma D')$, which is the tied velocity $-(1 - m^2)\partial_m L(m, \rho)/\rho$ at $\gamma = \rho$ (the $1/\rho$ is the chain rule through $w = u/\|u\|$, $\rho = \|u\|^2$; cf. Ren et al., 2025, Lemma B.1). Reparametrising time by $dt = dt'/\rho$ gives the claim.

(b) Write $z = m^{-(2k-2)}$. While $m \leq \bar{m}$ ($\bar{m} \downarrow m_0$ at the end), using $1 - m^2 \leq 1$ in the attraction, $1 - m^2 \geq 1 - \bar{m}^2$ and $V'(m)/m \geq 4k^2$ in the repulsion, dropping the term $-2k(1 - \frac{1}{N})\rho m^{2k-2}$, and using that D is increasing,

$$\frac{d \ln m}{dt} \leq s^2 \left[4km^{2k-2} - (1 - \bar{m}^2) \frac{4k^2\rho}{N} \right], \quad \dot{\rho} = 8s^2\rho(G - \rho D) \geq -8s^2D(\bar{m})\rho^2,$$

so $\rho \geq \rho_0/E$ with $E(t) = 1 + 8s^2D(\bar{m})\rho_0 t$, and $\dot{z} \geq -(2k-2)4ks^2 + (2k-2)(1 - \bar{m}^2)\frac{4k^2s^2}{N}\rho z$. Comparison with the linear equation gives $z(t) \geq (z_0 - K)E^a + KE$ with $K = k(k-1)/(\rho_0 D(\bar{m})(a-1))$ (and a evaluated at $\bar{m}$). By (3) $z_0 > K$ for $\bar{m}$ close to m_0 , so the right side is increasing in $E \geq 1$ and $z(t) \geq z_0$: $m \leq m_0 < \bar{m}$ for all t (continuity argument), $m \rightarrow 0$, and $\rho z \geq \rho_0(z_0 - K)E^{a-1} \rightarrow \infty$. Restarting the argument at a late time t_0 (where $ND(m(t_0)) \rightarrow 1$, so (3) holds with a arbitrarily close to $k^2(k-1)$) gives $m^{2k-2} \leq Ct^{-a'}$ for every $a' < k^2(k-1)$. Hence m^2 and G/ρ are integrable in t , so $\frac{d}{dt} \frac{1}{\rho} = 8s^2(D - G/\rho) = \frac{8s^2}{N} + (\text{integrable})$ gives $\frac{1}{\rho} = \frac{8s^2 t}{N} + c_1 + o(1)$. Then $\frac{d \ln m}{dt} = -\frac{4k^2s^2}{N}\rho(1 + O(m^2)) + O(m^{2k-2}) = -\frac{k^2}{2t} + (\text{integrable})$, so $m t^{k^2/2}$ converges to a positive limit.

(c) Write $q = \rho/\rho^*(m)$. Since $\partial_m L(m, \rho) = s^2\rho(\rho D' - 2G')$ and $\rho^* = 2G'/D'$, the tied flow is

$$\dot{m} = s^2(1 - m^2)2G'(1 - q) = 4ks^2(1 - m^2)m^{2k-1}(1 - q), \quad \dot{\rho} = 8s^2\rho(G - \rho D),$$

and $1/\rho^* = \frac{1}{2}(1 - \frac{1}{N}) + V'(m)/(4kNm^{2k-1})$; the first summand is the $O(\rho)$ term $2k(1 - \frac{1}{N})m^{2k-1}\rho$ of $\rho D'$, the second the noise term. As $V'(m)/m = \sum_{i \geq 1} 2i\beta_{2i}^2 m^{2i-2}$ is increasing from $4k^2$, for $m \leq \frac{1}{2}$

$$\rho \left[\frac{1}{2}(1 - \frac{1}{N}) + \frac{k}{Nm^{2k-2}} \right] \leq q \leq \rho \left[\frac{1}{2}(1 - \frac{1}{N}) + \bar{v}_k \frac{k}{Nm^{2k-2}} \right], \quad \text{and} \quad \rho^* < \frac{2}{1 - 1/N}. \quad (6)$$

Bootstrap. Let $T = \sup\{t \leq \tau_{\text{tied}} : q < \frac{1}{3} \text{ on } [0, t]\}$; $T > 0$ as $q(0) \leq \frac{1}{4}$. On $[0, T)$, $\dot{m} > 0$, so $m \geq m_0$, and $\dot{m} \geq 2ks^2m_0^{2k-1} > 0$ while $m \leq \frac{1}{2}$, so $\tau_{\text{tied}} < \infty$ if $T = \tau_{\text{tied}}$. With m as the variable on $[0, T)$,

$$\frac{d \ln \rho}{dm} = \frac{\dot{\rho}}{\rho \dot{m}} = \frac{2(G - \rho D)}{k(1 - m^2)m^{2k-1}(1 - q)} \leq \frac{2m}{k(1 - m^2)(1 - q)} \leq \frac{3m}{k(1 - m^2)},$$

and integrating from m_0 gives the upper bound in (4); with $m \leq \frac{1}{2}$ and $k \geq 2$, $\rho \leq (\frac{4}{3})^{3/4}\rho_0 < 1.241\rho_0$. By Lemma A.1 $\rho^*(m) \geq \rho^*(m_0)$, so $q \leq 1.241\rho_0/\rho^*(m_0) \leq 0.32 < \frac{1}{3}$ on $[0, T)$; by continuity $T = \tau_{\text{tied}}$, which proves the first two claims and the upper bound. *Lower bound.* Similarly

$$\frac{d}{dm} \frac{1}{\rho} = \frac{2(D - G/\rho)}{k(1 - m^2)m^{2k-1}(1 - q)}, \quad D - \frac{G}{\rho} = \frac{V}{N} - m^{2k} \left(\frac{1}{\rho} - 1 + \frac{1}{N} \right) \leq \frac{V}{N},$$

because $\rho \leq 0.32\rho^*(m_0) < 0.64/(1 - \frac{1}{N})$ by (6). With $1/(1 - q) \leq \frac{3}{2}$ and $V/(1 - m^2) \leq \frac{4}{3}V(\frac{1}{2})$ (both increasing), $\frac{d}{dm} \frac{1}{\rho} \leq \frac{4V(1/2)}{kNm^{2k-1}}$, and integrating from m_0 , $\frac{1}{\rho} \leq \frac{1}{\rho_0} + \frac{4V(1/2)}{k(2k-2)Nm_0^{2k-2}} = \frac{1}{\rho_0} + \frac{c_1}{Nm_0^{2k-2}}$. (The ratio ρ/ρ_0 is not $1 + O(\rho_0)$: as $\rho_0 \rightarrow 0$, $d \ln \rho / dm \rightarrow 2m/(k(1 - m^2))$, so $\rho(\tau_{\text{tied}})/\rho_0 \rightarrow (\frac{4}{3}(1 - m_0^2))^{1/k}$; the exponent in (4) is $1/(k(1 - \sup q))$ with $\sup q \leq \frac{1}{3}$. This growth affects τ_{tied} only through q .) *Escape time.* Since m is increasing, $\tau_{\text{tied}} = \int_{m_0}^{1/2} dm/\dot{m}$, i.e.

$$X\tau_{\text{tied}} = (2k - 2)m_0^{2k-2} \int_{m_0}^{1/2} \frac{m^{1-2k} dm}{(1 - m^2)(1 - q)}, \quad (2k - 2)m_0^{2k-2} \int_{m_0}^{1/2} m^{1-2k} dm = 1 - (2m_0)^{2k-2}.$$

Lower bound: use $\frac{1}{(1 - m^2)(1 - q)} \geq 1 + q$, the left inequality of (6) with $\rho \geq \rho_{\min}$, and the identity $(2k - 2)m_0^{2k-2} \int_{m_0}^{1/2} m^{3-4k} dm = (1 - (2m_0)^{4k-4})/(2m_0^{2k-2})$. Upper bound: for $m \leq \frac{1}{2}$, $q \leq \frac{1}{3}$, $\frac{1}{1 - m^2} \leq 1 + \frac{4}{3}m^2$ and $\frac{1}{1 - q} \leq 1 + \frac{3}{2}q$, so

$$\frac{1}{(1 - m^2)(1 - q)} \leq 1 + \frac{4}{3}m^2 + \frac{3}{2}q(1 + \frac{4}{3}m^2) \leq 1 + \frac{4}{3}m^2 + 2q;$$

the m^2 term integrates to $\frac{4}{3}\ell_k m_0^2$, and the right inequality of (6) with $\rho \leq \rho_{\max}$ gives $2q \leq \rho_{\max} + 2\bar{v}_k \rho_{\max} k / (Nm^{2k-2})$, whose two terms integrate to at most $\rho_{\max}$ and $\bar{v}_k \rho_{\max} k / (Nm_0^{2k-2})$. This is (5). Finally, the left side of (5) is $\geq 1 - (2m_0)^{2k-2} \geq 1 - 4m_0^2 \geq 1 - 4\lambda_k m_0^2$, and by (4) the right side is $\leq 1 + \frac{8}{3}\lambda_k m_0^2 + 1.25\rho_0 + (\frac{4}{3})^{3/(2k)}\bar{v}_k \varepsilon$ (using $\frac{4}{3}\ell_2 \leq \frac{8}{3}\lambda_2$, and $\frac{4}{3}\ell_k \leq \frac{4}{3} \cdot \frac{k-1}{k-2} \leq \frac{8}{3}$ for $k \geq 3$), which gives the bound on $|X\tau_{\text{tied}} - 1|$.

(iv) The per-prompt losses in a step are i.i.d. with mean L ; the gradient of their average has mean ∇L for every B (differentiation under $\mathbb{E}$ is justified by polynomial moments), and B enters only the covariance, which is $\propto 1/B$. $\square$

Remark 1 (The trap boundary is $c_k \rho^*$, not ρ^*). Condition (3) is sharp as $m_0 \rightarrow 0$: since $ND_0 \rightarrow 1$ and $a \rightarrow k^2(k - 1)$, it reads $\rho_0 > c_k Nm_0^{2k-2}/k(1 + o(1))$ with $c_k = 1 + 1/(k^3 - k^2 - 1)$ ($c_2 = \frac{4}{3}$, $c_3 = \frac{18}{17}$), and in the small- m system $\dot{m} = s^2 m(4km^{2k-2} - 4k^2\rho/N)$, $\dot{\rho} = -8s^2\rho^2/N$ (exactly $z = (z_0 - K)E^a + KE$ with $a = k^2(k - 1)$, $ND_0 = 1$) z reaches 0 in finite time, i.e. the flow escapes, for every $\rho_0 < c_k Nm_0^{2k-2}/k$. So for $\rho^*(m_0) < \rho_0 < c_k \rho^*(m_0)$ the overlap first decreases and then escapes: “ $\rho_0 > \rho^*(m_0) \Rightarrow \text{trap}$ ” is false. LSODA bisection of the exact flow gives the boundary $\rho_0/\rho^*(m_0) = 1.3333$ ($k=2$, $m_0=10^{-3}$), 1.0589 ($k=3$, $m_0=10^{-2}$) and 1.3735 at the E5 cell ($d=64$, $N=16$, $\rho^* = 0.1155$), where (3) gives $\rho_{\text{trap}} = 0.178$: $\rho_0 = 0.3$ is provably trapped, and $\rho_0 = 0.1 < \rho^*$ escapes. The escape half is proved only for the small- m system; for the exact flow it is a numerical statement.

Remark 2 (N -flatness of the tied readout is proved). Part (iii)(c) closes the gap between the frozen- ρ identity (iii)(a) and the trap (iii)(b): for $\rho_0 \leq \frac{1}{4}\rho^*(m_0)$ it bounds $\rho(t)$ on both sides up to $m = \frac{1}{2}$, and the tied escape time is $1/X$ up to a factor $1 + O(\varepsilon + \rho_0 + \lambda_k m_0^2)$ in which N enters only through $\varepsilon = k\rho_0/(Nm_0^{2k-2})$, so the N -flat third of the trichotomy is a theorem for $\varepsilon \ll 1$. When $\rho_0/(Nm_0^{2k-2}) = O(1)$ the readout can still decay substantially and the escape time is N -dependent: the E5 cell ($k = 2$, $\rho_0 = 0.1$, $N = 16$, $d = 64$) has $\rho_0/(Nm_0^2) = 0.4$ and $\rho_0/\rho^*(m_0) = 0.87$ (outside the hypothesis); along the flow ρ falls to $0.72\rho_0$, and the cell showed an N -ratio of 1.9 (flow: $\tau(16)/\tau(256) = 1.86$). `scripts/verify_tied_flat.py` integrates the exact flow on every tied cell of the paper: 10 of the 19 satisfy the hypothesis, and on them all bounds of (iii)(c) hold (80/80 checks; `results/verify_tied_flat.txt`).

Theorem B (Free readout, fixed N , $d \rightarrow \infty$). Assume (B1) $\sigma = \sigma_k$, $k \geq 2$; (B2) $N \geq 2$, $r > 0$, $\Gamma_0 > 0$, $s > 0$ and $\mu_0 > 0$ fixed; (B3) $m_0 = \mu_0 d^{-1/2}$, $d \rightarrow \infty$. Then

$$\tau_{\text{esc}} = (1 + o(1)) \exp\left(\frac{(4k - 2)\beta_2^2 \Gamma_0^2}{2r}\right) \frac{\mu_0^{-(4k-2)} d^{2k-1}}{4k(4k - 2)\alpha_k^8 s^2 N}, \quad \beta_2^2 = 2k^2, \alpha_k = 1.$$

In particular the escape exponent is $4k$ for every finite N , and N enters only as the prefactor $1/N$.

Proof. Rescale $t \mapsto s^2 t$ (this maps the flow with s to the flow with $s = 1$ and the same r), so $s = 1$. Put $m_1 = m_0 e^{-k^2 \Gamma_0^2 / r}$ and note $\frac{(4k-2)\beta_2^2 \Gamma_0^2}{2r} = (4k-2) \frac{k^2 \Gamma_0^2}{r}$, so the claim is $\tau_{\text{esc}} \sim \tau_B := m_1^{-(4k-2)} / (4k(4k-2)N)$. The equations are

$$\frac{d \ln m}{dt} = (1 - m^2) \left[4k \Gamma m^{2k-2} \left(1 - \frac{\Gamma}{2} \left(1 - \frac{1}{N} \right) \right) - \frac{\Gamma^2 V'(m)}{Nm} \right], \quad \dot{\Gamma} = -2rD(m)(\Gamma - \Gamma^*(m)),$$

with $V'(m)/m = 4k^2(1 + O(m^2))$, $D = N^{-1}(1 + O(m^2))$, $\Gamma^* = Nm^{2k}(1 + O(m^2 + Nm^{2k}))$; constants below depend on $(k, N, r, \Gamma_0, \mu_0)$ only.

Phase 1, $0 \leq t \leq t_1 = \frac{(k-1)N}{4r} \ln d$ (readout transient). While $m \leq 2m_0$, $\dot{\Gamma} = -\frac{2r}{N}(1 + O(d^{-1}))\Gamma + O(d^{-k})$, and $\Gamma \geq \Gamma_0 d^{-(k-1)/2}/2 \gg Nd^{-k}$, so $\Gamma(t) = \Gamma_0 e^{-2rt/N}(1 + o(1))$ uniformly. Integrating the m -equation, $\ln \frac{m(t_1)}{m_0} = -\frac{4k^2}{N} \int_0^{t_1} \Gamma^2(1 + o(1)) + O(m_0^{2k-2} \int (\Gamma + \Gamma^2)) = -\frac{k^2 \Gamma_0^2}{r} + o(1)$, because $\int_0^{t_1} \Gamma^2 = \frac{N\Gamma_0^2}{4r}(1 + o(1))$ and $m_0^{2k-2} \int (\Gamma + \Gamma^2) = O(d^{1-k}N/r)$; the same bounds keep $m \leq m_0(1 + o(1))$, closing the bootstrap.

Phase 2, $t_1 \leq t \leq t_2 = t_1 + \frac{(k+1)N}{2r} \ln d$ (relaxation onto Γ^*). Now $\Gamma \leq 2\Gamma_0 d^{-(k-1)/2}$, so $|\frac{d \ln m}{dt}| = O(d^{1-k}\Gamma + \Gamma^2)$ integrates to $o(1)$: $m(t_2) = m_1(1 + o(1))$. With $e = \Gamma - \Gamma^*$, $\dot{e} = -2rDe - \Gamma^* \dot{m}$ and $D \geq 1/N$, $|e(t_2)| \leq d^{-(k+1)}\Gamma_0 + \frac{N}{2r} \sup |\Gamma^* \dot{m}| = o(d^{-k}) = o(\Gamma^*(m(t_2)))$.

Phase 3, adiabatic, until t_3 , the first time $m = m_\delta$. Fix $\delta \in (0, \frac{1}{4})$ and choose m_δ (independent of d) so small that for $m \leq m_\delta$ the $O(\cdot)$ terms above are $\leq \delta$ and $\epsilon_\delta := 32k^2 Nm_\delta^{4k-2} \leq \delta r/N$. Write $\Gamma = \Gamma^*(1 + e')$. While $|e'| \leq 1$, $0 < \frac{d \ln m}{dt} = 4kNm^{4k-2}(1 + e')(1 + O(\delta))$ and, as $0 \leq \frac{d \ln \Gamma^*}{d \ln m} = 2k - \frac{mD'}{D} \leq 2k$, $|\frac{d \ln \Gamma^*}{dt}| \leq \epsilon_\delta$; and $\dot{e}' = -(2rD + \frac{d \ln \Gamma^*}{dt})e' - \frac{d \ln \Gamma^*}{dt}$. At $|e'| = \delta$ the right side has the sign of $-e'$, since $(\frac{2r}{N} - \epsilon_\delta)\delta > \epsilon_\delta$. Hence $|e'| \leq \delta$ on $[t_2, t_3]$, and $t_3 - t_2 = (1 + O(\delta)) \frac{m(t_2)^{-(4k-2)} - m_\delta^{-(4k-2)}}{4k(4k-2)N} = (1 + O(\delta) + o(1)) \tau_B$.

Phase 4, $t \geq t_3$. $(m, \Gamma)(t_3) \in K_\delta = \{m_\delta\} \times [(1 - \delta)\Gamma^*, (1 + \delta)\Gamma^*]$, a compact set on which $L < 1 = L(0, 0)$ (since $L = 1 - \Gamma G(2 - (1 + e'))$). L is a Lyapunov function, $\frac{dL}{dt} = -(1 - m^2)(\partial_m L)^2 - r(\partial_\Gamma L)^2$, trajectories are bounded, and the only equilibria in $[0, 1) \times [0, \infty)$ are $(0, 0)$ and, for $m \in (0, 1)$, points with $\Gamma = \Gamma^*(m)$ and $\frac{d}{dm} L(m, \Gamma^*(m)) = -R'(m) = 0$ (envelope theorem), which Lemma A.1 excludes. By LaSalle every trajectory from K_δ has $m \rightarrow 1$; the sets $\{x : m \text{ exceeds } \frac{1}{2} \text{ before time } T\}$ are open and cover K_δ , so the remaining time is $\leq T_\delta < \infty$, independent of d .

Phases 1, 2 and 4 last $O(N \ln d/r) + T_\delta = o(\tau_B)$, as $\tau_B \asymp d^{2k-1}$. Thus $\limsup |\tau_{\text{esc}}/\tau_B - 1| \leq C\delta$ for every δ . $\square$

Proposition 4 (Joint scaling; proof sketch). Let $N = \lambda d^k$, $r = \tilde{r}d$, $m_0 = \mu_0 d^{-1/2}$, $\Gamma_0 \in [0, 2)$, $s = 1$, and put $\mu = \sqrt{d}m$, $\theta = t d^{1-k}$. As $d \rightarrow \infty$ the flow converges, uniformly on compacts of $\{\mu > 0\}$ with an $O(1/d)$ error, to the d -free system

$$\frac{d\mu}{d\theta} = 4k\Gamma(1 - \frac{\Gamma}{2})\mu^{2k-1}, \quad \frac{d\Gamma}{d\theta} = 2\tilde{r}[\mu^{2k} - \Gamma(\mu^{2k} + \frac{1}{\lambda})], \quad (7)$$

whose solution blows up ($\mu \rightarrow \infty$) at a finite $T(\mu_0; \lambda, \tilde{r}, \Gamma_0)$, and $\tau_{\text{esc}} d^{1-k} \rightarrow T$. (The finite- N repulsion is $O(1/d)$ in this scaling.) With $\kappa := 2 - \partial \ln T / \partial \ln \mu_0$ (the secant exponent of the experiments is a finite-difference estimate of it): (a) $\lambda T \rightarrow \mu_0^{-(4k-2)} / (4k(4k-2))$ as $\lambda \rightarrow 0$ ($\tilde{r}, \Gamma_0, \mu_0$ fixed): $\kappa \rightarrow 4k$; (b) at $\lambda = \infty$ and fixed $\Gamma_0 \in (0, 2)$, $\mu_0^{2k-2} T \rightarrow 1 / ((2k-2)2k\Gamma_0(2 - \Gamma_0))$ as $\mu_0 \rightarrow 0$: $\kappa \rightarrow 2k$; (c) at $\lambda = \infty$, $\Gamma_0 = 0$ (readout growth-limited), $\mu_0^{2k-1} T \rightarrow \frac{\pi}{2} \frac{(2k-3)!!}{(2k-2)!!} / (2\sqrt{2k\tilde{r}})$ as $\sqrt{\tilde{r}}\mu_0 \rightarrow 0$: $\kappa \rightarrow 2k + 1$.

Sketch. Substituting, $d^{k-\frac{1}{2}} \dot{m} = (1 - \frac{\mu^2}{d})[2k\Gamma(2 - \Gamma(1 - \frac{1}{N}))\mu^{2k-1} - \frac{\Gamma^2}{\lambda} \sum_i 2i\beta_{2i}^2 \mu^{2i-1} d^{-i}]$ and $d^{k-1} \dot{\Gamma} = 2\tilde{r}[\mu^{2k} - \Gamma((1 - \frac{1}{N})\mu^{2k} + V(\mu d^{-1/2})/\lambda)]$, which is (7) up to $O((1 + \mu^{2k+2})/d)$. In (7), $\Gamma \leq \max(\Gamma_0, 1) < 2$ (at $\Gamma \geq 1$, $\dot{\Gamma} < 0$) and $\Gamma \geq c := \min(\Gamma_0, \Gamma_\infty(\mu_0))$ with $\Gamma_\infty = \mu^{2k}/(\mu^{2k} + \lambda^{-1})$ increasing along the (increasing) μ (for $\Gamma_0 = 0$, $\Gamma > 0$ for all $\theta > 0$ since $\dot{\Gamma} = 2\tilde{r}\mu^{2k} > 0$ there); so $\dot{\mu} \geq c'\mu^{2k-1}$ and μ blows up in finite time. Split at $\mu = M$: up to M , continuous dependence gives convergence of the hitting time; after M both the limit and (by the same invariance bounds, the repulsion being $O(M^{2-2k}/\lambda d)$) the finite- d flow need time $\leq CM^{2-2k}$, which $\rightarrow 0$ as $M \rightarrow \infty$. Corners: (a) Γ relaxes at rate $2\tilde{r}/\lambda \rightarrow \infty$ onto $\Gamma_\infty \approx \lambda\mu^{2k}$, the transient moves $\ln \mu$ by $O(\Gamma_0 \lambda / \tilde{r})$, and $\dot{\mu} = 4k\lambda\mu^{4k-1}(1 + o(1))$. (b) $d\Gamma/d\mu = \tilde{r}\mu(1 - \Gamma)/(2k\Gamma(1 - \Gamma/2))$, so $\Gamma = \Gamma_0 + O(\tilde{r}\mu_0^2 M^2)$ on $[\mu_0, M\mu_0]$, which carries all but $O(M^{2-2k})$

of T . (c) For $\Gamma \ll 1$, $\Gamma^2 = \frac{\tilde{r}}{2k}(\mu^2 - \mu_0^2)$ is an approximate first integral, so $\dot{\mu} = 4k\sqrt{\tilde{r}/2k}\sqrt{\mu^2 - \mu_0^2}\mu^{2k-1}$ and $T = \mu_0^{1-2k} \int_1^\infty \frac{dx}{4k\sqrt{\tilde{r}/2k}x^{2k-1}\sqrt{x^2-1}}$; the region where $\Gamma = O(1)$ costs $O(\tilde{r}^{k-1}) = o(\mu_0^{1-2k}\tilde{r}^{-1/2})$. The interchange of the corner limits with $\partial/\partial \ln \mu_0$ (needed for κ rather than for T) is not proved here; numerically the limits hold for the derivative as well (script). $\square$

Remark 3 (What is needed to transfer Theorem B to online SGD; heuristic). (1) Feature step. A drift $\eta c m^{\kappa-1}$ beats the spherical renormalisation drag $\asymp \eta^2 d \sigma_w^2 m/B$ at $m \asymp d^{-1/2}$ iff $\eta \lesssim cB d^{-\kappa/2}/\sigma_w^2$: the critical scaling $\eta \asymp d^{-\kappa/2}$ of Ben Arous et al. (2021), with $\kappa = 2k$ (pinned, tied) or $4k$ (free). Every w -gradient carries a factor Γ , so $\sigma_w^2 \asymp \Gamma^2/N$ near $m = 0$, which relaxes the free-readout requirement once $\Gamma \approx \Gamma^*$. (2) Readout noise. The SGD readout fluctuates around Γ^* with stationary variance $\text{Var}(\Gamma) \asymp \eta_\Gamma/B$ (relaxation rate and per-prompt variance both $\asymp 1/N$), independent of N ; as L is quadratic in Γ this adds $\text{Var}(\Gamma)D'(m)$ to the repulsion. Pathwise tracking, $\text{Var}(\Gamma) \ll \Gamma^{*2} \asymp N^2 d^{-2k}$ at $m \asymp d^{-1/2}$, needs $\eta_\Gamma \ll BN^2 d^{-2k}$; the mean m -drift alone needs only $\eta_\Gamma \ll BN^2 d^{1-2k}/k$. At $\eta_\Gamma = r\eta_0 d^{-2}$ and fixed N both fail as $d \rightarrow \infty$ ($k \geq 2$). In E2 ($k = 2, d = 64, \eta_0 = r = 1$) $d^2 = 4096$ against $BN^2 \geq 16384$; the E5 cell $\eta_\Gamma = 10\eta, N = 16$ violates it ($BN^2/r = 1638 < d^2$, constants aside), consistent with SGD being 24% slower than the flow there (2.26M vs 1.81M steps).

Conjecture 1 (SGD transfer). (a) Under the conditions of Theorem B, with $\eta = \eta_0 d^{-2k}$, $\eta_\Gamma = r\eta$ and $\eta_\Gamma \ll BN^2 d^{-2k}$ (i.e. $B \gg r\eta_0/N^2$), $\eta \cdot T_{0.5}/\tau_{\text{esc}} \rightarrow 1$ in probability. (b) In the setting of Theorem A(i) with $m_0 \asymp d^{-1/2} < m^*$, for every B and every $\eta \leq \eta_0 d^{-k}$, $\Pr(\max_{n \leq e^{cdm^2}} m_n \geq m^*/2) \rightarrow 0$; this would follow from a supermartingale bound for e^{adm^2} on $\{m \leq m^*\}$ (negative population drift plus spherical drag), in the manner of Ben Arous et al. (2021).

C. Experimental details

All toy experiments: float64 numpy, analytic gradients verified by finite differences (relative error $< 10^{-4}$), online spherical SGD with fresh data each step, $B = 32$ prompts per step unless stated, $\eta = \eta_0/d^2$. Projected samplers (exact in distribution) were used where the gradient depends on x only through a few projections. Compute: Exp. 1 6.4, Exp. 1b 2.8, Exp. 2 3.2, Exp. 3 2.3, Exp. 4 6.0, Exp. 5 6.0 CPU-hours on 4 cores.

Exp. 1 (single skill, uniform initialisation). $k^* \in \{1, 2\}$, $d \in \{8, \dots, 128\}$, 5 seeds, $N = 128$, fixed $\gamma = 0.1$ (A) and $a = 1$ (B). Slopes of $\log T_{0.5}$ vs $\log d$: B $k^* = 1$: 1.98 ± 0.06 ($d \geq 16$; the pre-registered 1.5 was an error: for $\kappa < 2$ the drift is $O(1)$ and $T \propto 1/\eta = d^2$); B $k^* = 2$: 2.33 ± 0.22 ; A $k^* = 1$: 2.71 ± 0.43 ; A $k^* = 2$: $3.1\text{--}3.5$ (20 of 105 runs censored at $3 \cdot 10^6$ steps). Regression of $\log(T_{0.5}\eta)$ on $\log|m_0|$: A $k^* = 2$: -2.19 ± 0.08 (reached runs with $\eta_0 \in \{1, 2\}$, $d \geq 16$; -2.29 ± 0.07 on all reached runs), i.e. $\kappa \approx 4.2\text{--}4.3$, a lower bound because of censoring; B $k^* = 2$: -0.57 ± 0.06 (logarithmic, $\kappa = 2$); A $k^* = 1$: -0.50 ± 0.06 ; B $k^* = 1$: -0.13 ± 0.02 .

Exp. 1b (fixed m_0). Threshold grid: $\gamma \in \{1, 0.1\}$, $N \in \{32, 128, 512\}$, $d \in \{16, 32, 64\}$, 3 seeds, $3 \cdot 10^5$ steps. Escapes at $\gamma = 1$: $N = 32$: 2/3, 0/3, 0/3; $N = 128$: 3/3, 1/3, 0/3; $N = 512$: 3/3, 3/3, 3/3 (corrected $d^* = 6.8, 30.8, 126.8$). $\gamma = 0.1$: 27/27. P7: $\gamma = 1$, $N = 128$, $d = 64$, $B \in \{32, 256, 1024\}$ with $3 \cdot 10^5, 10^5, 3 \cdot 10^4$ steps: 0/9 escapes, final $|m|$ 0.005–0.015 ($B \leq 256$) and 0.063–0.067 ($B = 1024$). Exponent table ($d = 32$, 4 values of m_0 , 3 seeds, $\eta = 1/d^2$, $\Gamma_0 = 0.01$ or $\rho_0 = 0.01$): in Fig. 2; censored runs ($N = 32$: 3/12 for both free protocols; $N = 128$, 10η : 2/12) are all at the smallest m_0 .

Censored fits and absolute-time ratios. For the three E1 cells containing censored runs (all at the smallest m_0) we refit $\log(T_{0.5}\eta) = a + b \log m_0$ by a Tobit likelihood with right-censoring at the step cap (‘results/README_stats.md’): κ_{eff} becomes 6.05 ± 0.18 (free $\eta_\Gamma = \eta$, $N = 32$; ODE 6.47), 6.73 ± 0.18 (free 10η , $N = 32$; ODE 7.02) and 6.67 ± 0.37 (free 10η , $N = 128$; ODE 6.18); the other nine cells are unchanged. All twelve are within 0.49 of the ODE and ten within 2 s.e.; bootstrap s.e. in the censored cells are 0.21–0.49. The Tobit fit assumes censored runs would eventually cross $m = \frac{1}{2}$, which is an assumption (their final $|m|$ is 0.04–0.11). Absolute-time ratios median($T_{0.5}$)/ODE: E1 (45 reached cells, ODE at each run’s exact m_0): 24/35/45 within 10/20/50%, range 0.64–1.34; E2 (18 cells, pinned $\gamma = 1$ excluded as the ODE predicts a trap): 17/18/18, range 0.85–1.10. Single seeds range 0.50–2.08 $\times$. One censored E1 cell (free 10η , $N = 128$, $m_0 = 0.5d^{-1/2}$: 1/3 reached) is at least 2.35 $\times$ slower than the ODE.

Exp. 6 (three regimes). $d = 64$, $m_0 = d^{-1/2}$, $\eta = 1/d^2$, $N \in \{16, 64, 256\}$, schemes $NB = 4096$ and $B = 64$, protocols pinned $\gamma \in \{1, 0.1\}$, free ($\Gamma_0 = 0.01$, $\eta_\Gamma = \eta$), tied ($\rho_0 = 0.01$), 3 seeds, 10^6 steps ($2.5 \cdot 10^6$ for the free $N = 16$

rerun). ODE predictions were computed and committed before the run. 12 of 72 runs are duplicates ($N = 64$ has $B = 64$ in both schemes). Compute 2.4 CPU-h. Medians of three seeds resolve escape times to roughly 5%. ODE values that moved by more than 1% between the Euler and LSODA integrators were updated to LSODA ('scripts/ode_lsoda.py'); values within 1% are quoted as originally computed; the SGD/ODE ratio statistics above were computed with the earlier Euler values and would rise by 0.6–3.6%.

Exp. 7 (controls, E5). As Exp. 6 with $B = 64$, 3 seeds, cap $2.5 \cdot 10^6$ (10^6 for the tied $\rho_0 = 0.3$ cell): free $\Gamma_0 = 0.01$, $\eta_\Gamma = 10\eta$: 2.26M/498k/212k (ODE 1.81M/500k/205k); free $\Gamma_0 = 0.1$, $\eta_\Gamma = \eta$: 1.52M/271k/166k (ODE 1.49M/270k/161k); tied $\rho_0 = 0.1$: 37k/18k/17k (ODE 33k/19k/17k; one $N = 16$ seed at 567k); tied $\rho_0 = 0.3$: 716k/27k/21k (ODE: trap/31k/21k); pinned $\gamma = 0.01$: 1.71M/1.64M/1.62M (ODE 1.70M/1.64M/1.63M); $\eta \propto B$ at pinned $\gamma = 0.1$, $N = 64$: 773k/192k/48k steps for $B = 16/64/256$, flow time 47.2/47.0/46.9. Compute 2.2 CPU-h.

Exp. 2 (many skills). $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, $\Gamma = 0.1$, $N = 128$, 3 seeds (plus 2 extra and $\alpha = 1$). Skills 1–8 learned in every run; emergence of skills 9–16 partial. In-weight baselines: (i) readout $\rho = \sum_p a_p/M$ (the readout-1 baseline does not learn); (ii) the 2-homogeneous student $\sum_j \|u_j\|^2 \sigma_2(\hat{u}_j \cdot x)$, $\|u_j\|_0^2 = 0.01$, $\eta \in \{0.25, 1, 4\}/d^2$, 3 seeds each: slope 0.75 ± 0.09 (1000-step log), $0.87\text{--}0.89 \pm 0.09$ (100-step log), $T_p \eta a_p = 0.23\text{--}0.26$; 18.7, 10.7, 9.0, 6.3, 2.7, 3.0, 2.7, 3.7 features on skills 1–8, best single alignment $0.86 \rightarrow 0.59$, pooled alignment 1.000 for all 16 skills, population MSE $3 \cdot 10^{-8}$: collective fit. Both are collective at $M = 64 > d$; a decoupled in-weight regime would need $M < d$ (as in Ben Arous et al. (2025), whose plateau length $\log(d/r_s)$ comes from the Gram matrix $WW^\top$ having eigenvalues $\approx r_s/d$ at initialisation—at $M \geq d$ that matrix is already full rank, which is our reading of why no search plateau and no single-feature emergence appear) and was not run. Slopes and exponents as in Sec. 4; the $\alpha = 1$ slope is 0.88 ± 0.09 (A) vs 0.61 ± 0.08 (B).

Exp. 3 (composition). $P = 2$, $\pi = (0.75, 0.25)$, $M = 4$, $\Gamma = 0.1$, $N = 128$, 5 pre-registered seeds + 5 extra + 3 at $\Gamma = 0.3$. Additivity residual with a paired estimator (same inputs, 17 queries per context): $\max |R| \leq 0.0086$ over 13 runs (the pre-registered independent-set estimator was too noisy, 0.04–0.35). Final neuron assignments (skill 1, skill 2): (4, 0), (2, 2), (2, 2), (2, 1), (2, 2) for the pre-registered seeds. Neuron arrival times $T \approx K/(\eta \pi_p m_0^2)$ with $K = 0.60$ (0.54–0.78).

Exps. 4–5 (transformers). Tokens $[x_i; y_i]$, query $[x_q; 0]$; embedding Linear($d+1 \rightarrow 64$), 2 pre-LN blocks (4-head causal softmax attention + ReLU MLP), linear readout; Adam 10^{-3} , fp32, online data, eval every 200 steps on 2048 held-out prompts, T_e = first eval with normalised MSE < 0.5 . Exp. 4: MLP width 256, $d \in \{16, 32\}$, $N \cdot B = 8192$. Exp. 5: $d = 256$, width 32, cells (a) $N=256, B=64$: $T_e = 4000, 2800$, third seed censored at 17k on the plateau; (c) $N=16, B=64$: none in 30k (3/3); (d) $N=16, B=1024$: 14800, 10600, 11600; (e) $k^* = 1$, $N=16, B=64$: 1400, 1200; (f) in-weight: 400, 200; initial $\max_j |\cos(W_j, v)|$: 0.126, 0.152, 0.176. Exp. 9 ($k^* = 3$, single neuron, sweep9.py): $\eta = 2 \cdot 10^{-4}$ for all $d, B = 64$, $m_0 = d^{-1/2}$ exactly, cap 10^6 steps, 3 seeds, 27 runs, 0.89 CPU-h; medians/ODE tied $N=128$: 1.18, 0.89, 0.80 ($d = 8, 16, 32$), pinned $\gamma=0.2$ $d=8$: 1.02; post-hoc controls: $d=8$ at $\eta/4$: 1.007; $d=32$ at $B=256$: 1.14 (declared before running). Exp. 8 (transformer_exp8.py, same model/data code as Exp. 5): readout frozen at its initial norm (0.56–0.58) or at $0.1 \times$ it, or trainable; cells $N=16, B=64$; $N=16, B=1024$; $N=256, B=64$; 2 seeds; 30k steps; 5.5 CPU-h; one pinned-small $N=256$ seed censored at 17k (budget). Exp. 11 (transformer_exp11.py): attn.out of both blocks frozen at init (norms 4.6) or at $0.1 \times$ (norms 0.46), same cells, 2 seeds, 30k steps, 6.0 CPU-h; one $0.1 \times$ cell has one seed, two $0.1 \times$ runs censored on the plateau at 8k and 10.8k (budget). Exp. 10 (sweep10.py): E2's protocols at $d = 128$, $\eta = 1/d^2$, $B = 64$; pinned $\gamma=1$ (2 seeds, cap 10^6), pinned $\gamma=0.1$ (3 seeds, cap $2.5 \cdot 10^6$), tied $\rho_0=0.01$ (3 seeds), free $\Gamma_0=0.01$, $\eta_\Gamma=\eta$ at $N=256$ (2 seeds); 26 runs, 3.8 CPU-h.

D. Pre-registration summary

All predictions were written to a version-controlled document before the corresponding run; the full text with timestamps, outcomes and changelog is in the repository (docs/preregistration.md). Summary:

Context length and readout parameterisation select the emergence exponent of in-context skills

ID	prediction	outcome
P1	MC drift of A, $k^*=2$, $\propto m^3$; closed form within 20%	met (slope 3.14 ± 0.07 ; $\leq 3\%$ at $m \geq 0.2$)
P2	$\gamma=1$, $N=128$: stuck for $d \geq 32$	partly ($d=64$ stuck, $d=32$: 1/3 escaped); threshold formula later corrected, $d^* \approx 31$
P3	$\gamma=0.1$: escapes at all d	met (27/27)
P4	free readout: κ rises as N falls, $4 \rightarrow 8$	direction met; magnitudes match the ODE at $d=32$ ($4.85 \rightarrow 7.0$), not the asymptotes
P4'	tied readout: $\kappa \approx 4$ at all N ; $\rho_0=1$ stuck for $d \geq 32$ (flow traps; SGD delays)	κ met (4.10–4.44); $\rho_0=1$: $d=32$ escaped 3/3, $d=64$ stuck
P5	learned $\Gamma \approx \Gamma^*(0.5; N)$	not met (relaxation too slow at $\eta_\Gamma = \eta$)
P6	A vs B slope difference ≈ 1 at $k^*=2$	met via m_0 -regression (κ 4.2 vs 2); d -slopes 0.9 ± 0.5
P7	no escape at any B	met (0/9)
P8	transformer, $d=32$: T_e falls $\geq 3\times$ from $N=16$ to 256 at fixed tokens	failed, reversed (1000 $\rightarrow$ 2800)
P9	$k^*=1$ weak N -dependence; in-weight fastest	partly ($\times 2.0$; in-weight fastest)
P10	N -dependence in both softmax and linear attention	met (same direction)
P11a–b	$d=256$, width 32: (c) stuck, (d) stuck	(c) stuck 3/3; (d) emerged 3/3 at $\approx 3\times$ (a): both rival and default failed
P11c	$k^*=1$ and in-weight controls fast	met
P12	additivity residual $ R \leq 0.02$	failed as pre-registered (estimator noise); identity confirmed with a paired estimator
P13	no extra compositional delay	supported with substitute statistic (floor made the original unreachable)
P14	multiplicative accuracy, Gaussian closed form	approximately multiplicative; Gaussian form wrong
P15	product composition not learned	met
P19	$d=64$: pinned-1 trapped; free $T \propto \text{ODE}$ with $T(16)/T(256) = 4.4$; tied flat within 5%	pinned-1: 0/9 at $N \leq 64$, 3/6 at $N=256$; free within one s.e. of ODE, ratio 4.4; tied flat within $\times 1.25$ (5% not resolvable)
P20	controls: ratios 8.9/9.1 (free $10\eta/\Gamma_0=0.1$), tied $\rho_0=0.1$ ratio 1.9, tied $\rho_0=0.3$ traps at $N=16$, pinned 0.01 flat, $\eta \propto B$ flow-invariant	10.7/9.2; 2.15; escaped after 30$\times$ delay (failed as worded) ; 1.06; invariant to 0.5%
P21	transformer readout protocols: pinned $O(1) \Rightarrow N=16$ stuck at $B=1024$; pinned small $\Rightarrow N=16$, $B=64$ emerges; free reproduces Exp. 5	failed (8400, 5800 at $B=1024$); failed (stuck 2/2); held ($3.7\times$, $3.8\times$)—kill criterion met
P24	transformer attention-output scale: frozen $O(1) \Rightarrow N=16$ stuck at $B=1024$; frozen $0.1\times \Rightarrow N=16$, $B=64$ emerges; $O(1)$ within $1.5\times$ of free at $N=256$	failed (15.0k, 26.4k); failed (stuck 2/2); failed ($3.4\times$, $2.6\times$ slower)—kill criterion met
P22	$k^*=3$: tied $\rho_0=0.01$ secants 6.5/6.4, medians within 15%; tied N -ratio 1.16/1.63/ ∞ ($d=8/16/32$); pinned $\gamma=0.2$: $d=8$ escapes, 16, 32 trapped	secants 5.7/6.0 (kill not triggered); medians 1.18/0.89/0.80 (failed ; controls : 1.007 at $\eta/4$, 1.14 at $B=256$); 1.14/19/ ∞ (failed at $d=16$: 47k/500k/censored); held (1.02; 3/3 censored twice)
P23	$d=128$: pinned $\gamma=1$ trapped at all N ; pinned 0.1: $N=16$ trapped, $N=64/256$ at $1.80/1.46 \cdot 10^6$; tied flat (ratio 1.09); free $N=256$ at $5.37 \cdot 10^6$, secant 7.4	held; held (3/3 censored; 0.99, 1.01); held (1.12/0.97/1.02; ratio 1.20); median held (0.985), secant 5.4 (pre-registered baseline was the wrong cell; like-for-like flow 5.4)
P16	T_p slopes 1.5 (A), 0.75 (B)	met (1.44–1.53; B 0.75 capacity-matched, 0.75–0.89 2-homogeneous)
P17	A steps $\geq 2\times$ sharper	met ($3.8\times$)
P18	loss exponents 0.33 (A), 0.67 (B)	A within tolerance (0.40); B failed (collective fit saturates)

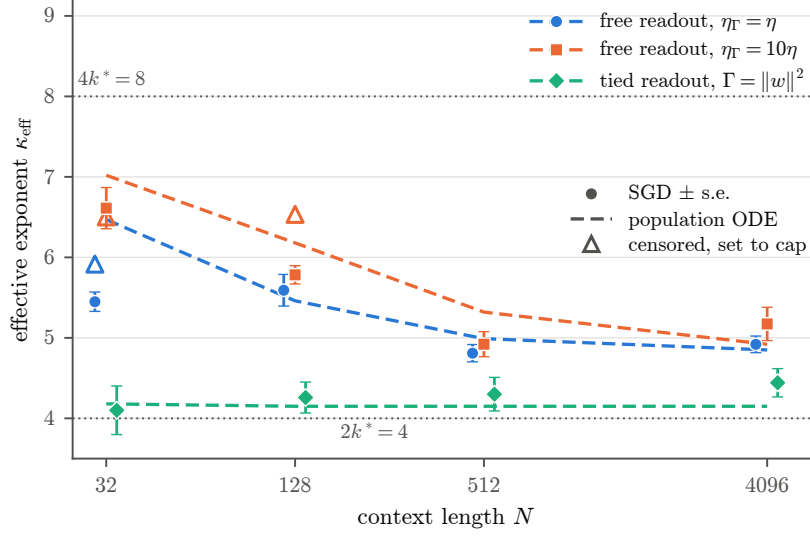


Figure 2. Effective exponent of skill acquisition for $k^* = 2$ at $d = 32$: online SGD (points, $\pm$ s.e.; triangles: censored runs set to the step cap) against the parameter-free population ODE (dashed). The tied readout stays near $2k^* = 4$ (4.1–4.4); the free readout rises towards $4k^* = 8$ as the context shortens.

E. Figures moved from the main text

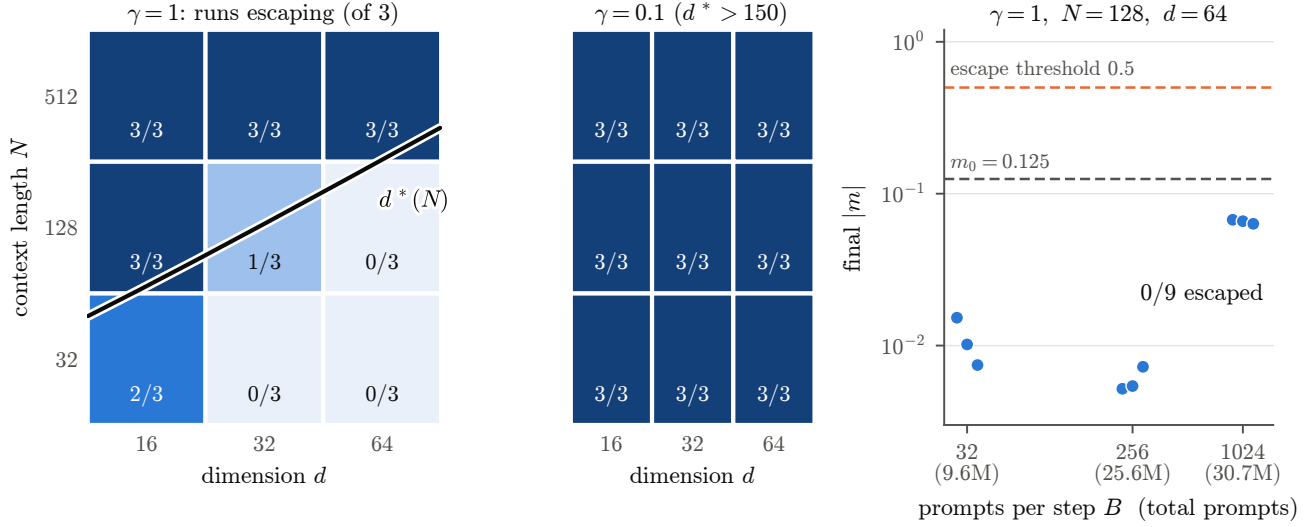


Figure 3. Left: fraction of runs escaping $m_0 = d^{-1/2}$ within $3 \cdot 10^5$ steps (fixed readout, 3 seeds per cell; curve: $d^*(N)$ of Prop. 2; for $\gamma = 0.1$ the curve lies above the grid). Right: at $\gamma = 1$, $N = 128$, $d = 64$ no run escapes for $B \in \{32, 256, 1024\}$ prompts per step (total prompts $\times 1, \times 2.7, \times 3.2$) and the final overlap falls below m_0 : more prompts do not substitute for context length.

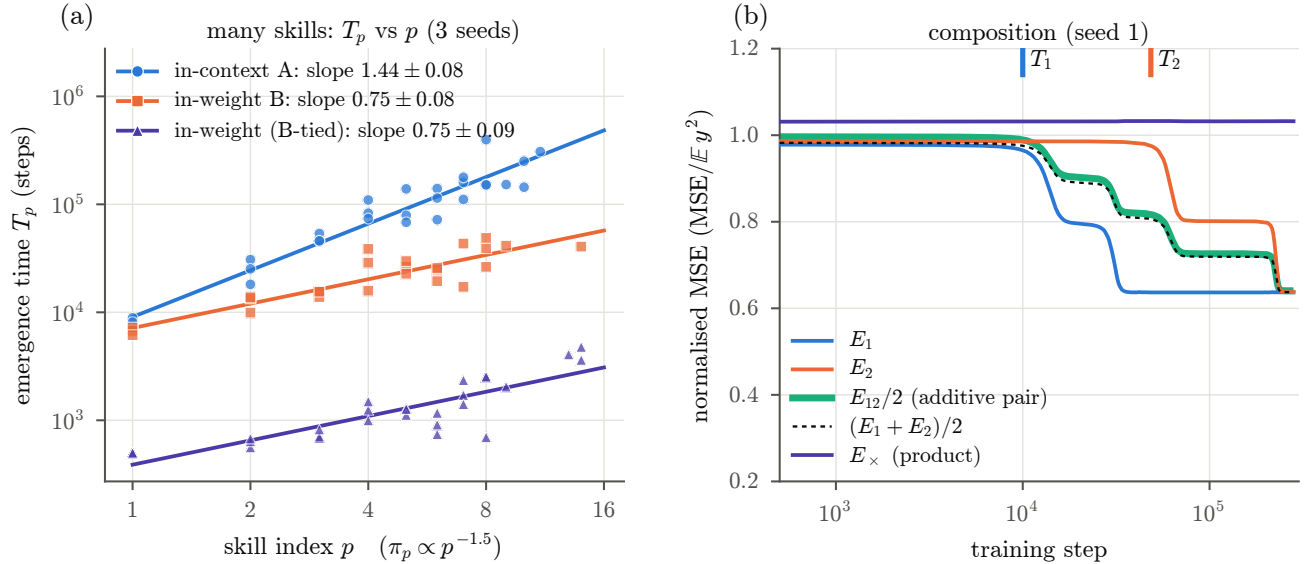


Figure 4. (a) Emergence time (drop midpoint) of skill p versus skill index for the in-context model (slope 1.44 ± 0.08), the capacity-matched in-weight baseline (slope 0.75 ± 0.08) and the 2-homogeneous in-weight student (slope $0.75\text{--}0.89$), $\pi_p \propto p^{-1.5}$, three seeds each; unlearned skills omitted. (b) Composition (Exp. 3, one seed): normalised MSE on skill-1 prompts, skill-2 prompts, the additive pair (halved) and the product pair; the dotted curve is half the sum of the single-skill losses and lies on the pair curve. Plateaus at 0.64 are the fixed-readout floor $(1 - \Gamma n)^2$, not zero.

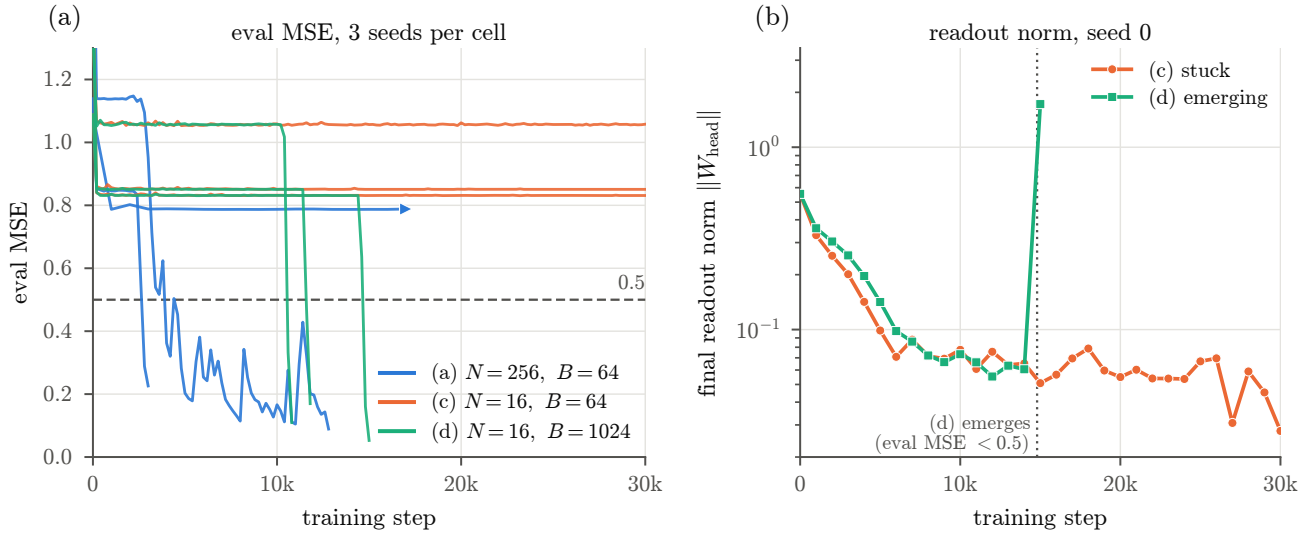


Figure 5. Two-layer softmax transformer, $d = 256$, MLP width 32, $k^* = 2$. (a) Evaluation MSE for $N=256, B=64$; $N=16, B=64$; and $N=16, B=1024$ (same tokens per step as the first), three seeds each; one $N=256$ seed was censored at 17k steps on the plateau. (b) Norm of the final readout for one stuck run ($N=16, B=64$) and one emerging run ($N=16, B=1024$, same seed): the norm collapses while the skill is unlearned; the regrowth at emergence is a single logged point (norms logged every 1000 steps).